The sharp A_p bound for the Hardy–Littlewood maximal operator

An exposition of Lerner's proof of Buckley's theorem

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Abstract

In 1972, B. Muckenhoupt [5] proved that the Hardy–Littlewood maximal operator is bounded on the weighted space $L^p_w(\mathbb{R}^d)$, $1 < p < \infty$, if and only if w satisfies the A_p condition. In 1993, S. Buckley [1] showed that its operator norm is bounded by a constant times $[w]_{A_p}^{1/(p-1)}$, and that this exponent is best possible. In 2008, A. K. Lerner [4] gave an alternative proof of Buckley's bound. We present Lerner's proof in detail, together with Buckley's example showing that the exponent cannot be improved.

1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$ we write $w(E) := \int_E w(x)\,dx$.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 \leq p < \infty$. We define*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \geq 0 : w(\{|f| > t\}) = 0\} < \infty$.

The central object of this report is the Hardy–Littlewood maximal operator M. Our proofs also use its analogue with respect to the measure $w(x)\,dx$.

Definition 3 (Hardy–Littlewood maximal operator)**.** *Let w be a weight with $w(Q) > 0$ for every cube Q. For a measurable function f and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)|\, w(y)\,dy,$$

where the supremum is taken over all closed cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing x. All cubes in this report are of this kind. If the supremum is taken only over the cubes $Q(x,r) := \{y : |y - x|_\infty \leq r/2\}$, $r > 0$, where $|y|_\infty := \max_{1 \leq k \leq d} |y_k|$, we write M^c_w. If $w \equiv 1$ we drop the subscript and write M and M^c.

It is well known that M is of weak type $(1,1)$ and strong type (p,p) for $1 < p \leq \infty$ [3, Theorem 1.1]. For $1 < p < \infty$ and a weight w we say that M is bounded on $L^p_w(\mathbb{R}^d)$ if there is a constant C such that

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p \, w \, dx \right)^{1/p} \leq C \left(\int_{\mathbb{R}^d} |f|^p \, w \, dx \right)^{1/p} = C \, \|f\|_{L^p_w} \tag{1}$$

for all $f \in L^p_w(\mathbb{R}^d)$.

Remark 1 (The zero weight)**.** *If $w = 0$ almost everywhere, then $\|f\|_{L^p_w} = 0$ for every f, so $L^p_w(\mathbb{R}^d)$ is trivial and M is trivially bounded.*

Remark 2 (The endpoint $p = \infty$)**.** *For $p = \infty$, the operator M is bounded on $L^\infty_w(\mathbb{R}^d)$ if and only if $w > 0$ almost everywhere or $w = 0$ almost everywhere. Muckenhoupt [5, Theorem 3] proves this on an interval, and the same argument works in $\mathbb{R}^d$. This is why we only consider $1 < p < \infty$.*

Suppose that (1) holds and let Q be a cube. Since $Mf(x)$ is at least the average of $|f|$ over Q for $x \in Q$, writing out the norms in (1) by Definition 2 gives

$$\left(\frac{1}{|Q|} \int_Q |f| \, dx \right)^p w(Q) \leq \int_Q (Mf)^p \, w \, dx \leq C^p \int_{\mathbb{R}^d} |f|^p \, w \, dx$$

for every $f \in L^p_w(\mathbb{R}^d)$. Each choice of f thus gives a condition on w, and the choice $f = w^{-\frac{1}{p-1}} \chi_Q$ makes this condition as strong as possible. The following proposition makes this precise, and we prove it in Appendix A.

Proposition 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (1) holds with constant C, then for every cube Q we have $\int_Q w^{-\frac{1}{p-1}} \, dx < \infty$ and*

$$\left(\frac{1}{|Q|} \int_Q w \, dx \right) \left(\frac{1}{|Q|} \int_Q w^{-\frac{1}{p-1}} \, dx \right)^{p-1} \leq C^p. \tag{2}$$

In particular, $w > 0$ almost everywhere.

This motivates the following definition [5, (7.10)].

Definition 4 (Muckenhoupt A_p weights)**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|} \int_Q w(x) \, dx \right) \left(\frac{1}{|Q|} \int_Q w(x)^{-\frac{1}{p-1}} \, dx \right)^{p-1}, \tag{3}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$. We say that $w \in A_p$ if $[w]_{A_p} < \infty$.

Definition 5 (Dual weight)**.** *Let w be a weight and let $1 < p < \infty$. The* dual weight *of w is $\sigma := w^{-\frac{1}{p-1}}$, with $\sigma = \infty$ where $w = 0$ [1, Section 1]. For measurable sets E we write $\sigma(E) := \int_E \sigma \, dx$.*

With this notation (3) takes the more compact form

$$[w]_{A_p} = \sup_Q \frac{w(Q)}{|Q|} \left(\frac{\sigma(Q)}{|Q|} \right)^{p-1}.$$

Taking the supremum over all cubes in Proposition 1 gives the following.

Corollary 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (1) holds with constant C, then $w \in A_p$ and*

$$[w]_{A_p} \leq C^p. \tag{4}$$

Muckenhoupt proved that the condition $w \in A_p$ is also sufficient.

Theorem A (Muckenhoupt [5, Theorem 9])**.** *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. Then M is bounded on $L^p_w(\mathbb{R}^d)$ if and only if $w \in A_p$.*

Theorem A shows that the operator norm of M on $L^p_w(\mathbb{R}^d)$, that is, the smallest constant C in (1), is finite precisely when $w \in A_p$. By Corollary 1, this norm is bounded from below by $[w]_{A_p}^{1/p}$. The natural question is therefore whether it is also bounded from above by a power of $[w]_{A_p}$, and which power is the smallest possible. This was answered by Buckley [1, Theorem 2.5].

Theorem 1 (Buckley)**.** *Let $1 < p < \infty$. There is a constant $c_{p,d}$, depending only on p and d, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,*

$$\|Mf\|_{L^p_w} \leq c_{p,d}\, [w]_{A_p}^{\frac{1}{p-1}}\, \|f\|_{L^p_w}.$$

Proposition 2 (Buckley)**.** *Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$. Then there is no constant c such that $\|Mf\|_{L^p_w} \leq c\, [w]_{A_p}^{\alpha}\, \|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.*

Buckley's proof of Theorem 1 [1, proof of Theorem 2.5], like Muckenhoupt's proof of Theorem A [5, Lemma 5 and Theorem 2], relies on a deep property that every A_p weight belongs to $A_{p-\varepsilon}$ for some $\varepsilon > 0$ [3, Theorem 4.5], together with interpolation. Lerner [4, Theorem C] found a short and elementary proof of Theorem 1. We present his argument in detail and complete it with a proof of Proposition 2, based on Buckley's example in $\mathbb{R}$ [1, proof of Theorem 2.5], which we extend to $\mathbb{R}^d$.

Remark 3 (An alternative proof of Theorem A)**.** *If M is bounded on $L^p_w(\mathbb{R}^d)$, then $w \in A_p$ by Corollary 1, which follows from Proposition 1. Conversely, if $w \in A_p$, then M is bounded on $L^p_w(\mathbb{R}^d)$ by Theorem 1. Combined, these results imply Theorem A.*

2 Proofs

2.1 Preliminaries

Lemma 1. *Let Q be a cube with side length $\ell(Q)$ and let $y \in Q$. Then $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, where $3Q$ denotes the cube with the same centre as Q and side length $3\ell(Q)$.*

Proof. Every $z \in Q$ satisfies $|z - y|_\infty \leq \ell(Q)$, since both z and y lie in Q. Hence $Q \subseteq Q(y, 2\ell(Q))$. If c is the centre of Q, then $|y - c|_\infty \leq \frac{1}{2}\ell(Q)$, so every $z \in Q(y, 2\ell(Q))$ satisfies $|z - c|_\infty \leq |z - y|_\infty + |y - c|_\infty \leq \frac{3}{2}\ell(Q)$, that is, $z \in 3Q$. □

Lemma 2. *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $Mf(x) \leq 2^d M^c f(x)$ for every $x \in \mathbb{R}^d$.*

Proof. Let $x \in \mathbb{R}^d$. For a cube Q containing x, Lemma 1 gives $Q \subseteq Q(x, 2\ell(Q))$, and since $|Q(x, 2\ell(Q))| = 2^d|Q|$ we obtain

$$\begin{aligned} Mf(x) &= \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f(y)|\, dy \\ &\leq \sup_{Q \ni x} \frac{2^d}{|Q(x, 2\ell(Q))|} \int_{Q(x,2\ell(Q))} |f(y)|\, dy \\ &\leq 2^d \sup_{r>0} \frac{1}{|Q(x,r)|} \int_{Q(x,r)} |f(y)|\, dy \\ &= 2^d M^c f(x). \end{aligned}$$

□

Lemma 3. *Let $1 < p < \infty$ and let $w \in A_p$ be a weight that is not almost everywhere zero. Then $w > 0$ almost everywhere and σ, as in Definition 5, is a weight. Moreover $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q.*

Proof. Let Q be a cube with $w(Q) > 0$. By the definition of $[w]_{A_p}$,

$$\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1} \leq [w]_{A_p} \frac{|Q|}{w(Q)} < \infty, \tag{5}$$

so $\sigma(Q) < \infty$ on any cube Q where $w(Q) > 0$.

Assume, for contradiction, that $w = 0$ on a set E with $|E| > 0$. Since w is not almost everywhere zero, $w > 0$ on a set F with $|F| > 0$, so we can choose a cube Q with $|Q \cap E| > 0$ and $|Q \cap F| > 0$. Then $w(Q) \geq w(Q \cap F) > 0$, so $\sigma(Q) < \infty$ by (5). But $\sigma = \infty$ on $Q \cap E$ by Definition 5, so $\sigma(Q) = \infty$, a contradiction. Therefore $w > 0$ almost everywhere. In particular $w(Q) > 0$, and hence $\sigma(Q) < \infty$ by (5), for every cube Q.

Since $w(Q) < \infty$ for every cube Q, we have $w < \infty$ almost everywhere, so $\sigma > 0$ almost everywhere and $\sigma(Q) > 0$ for every cube Q. As every compact set lies in a cube, σ is locally integrable and hence a weight. □

Remark 4 (The Vitali covering lemma)**.** *In the proof of the weak type $(1,1)$ estimate for M [3, Theorem 1.1], a compact set $K \subseteq \{Mf > \lambda\}$ is covered by finitely many cubes Q with $\frac{1}{|Q|}\int_Q |f|\, dx > \lambda$. The Vitali covering lemma [3, Lemma A.4], which holds for balls as well as cubes, selects disjoint cubes $Q_1, \ldots, Q_N$ among them with $K \subseteq \bigcup_j 3Q_j$. Since $|3Q_j| = 3^d|Q_j|$, this gives*

$$|K| \leq \sum_{j=1}^N |3Q_j| = 3^d \sum_{j=1}^N |Q_j| \leq \frac{3^d}{\lambda} \int_{\mathbb{R}^d} |f|\, dx.$$

For the measure $v(x)\, dx$, the equality $|3Q_j| = 3^d|Q_j|$ would have to be replaced by an inequality $v(3Q_j) \leq C\, v(Q_j)$, and no constant C works for all weights v. For example, if $v = \varepsilon$ on Q_j and $v = 1$ on the rest of $3Q_j$, then $v(3Q_j)/v(Q_j) \geq (3^d - 1)/\varepsilon$, which is arbitrarily large. The Besicovitch covering lemma covers K by the selected cubes themselves, without enlarging them, and therefore gives a constant independent of v.

Lemma 4 (Besicovitch covering lemma [2, Theorem 1.1])**.** *Let $E \subset \mathbb{R}^d$ be a bounded set, and suppose that for each $x \in E$ a cube $Q(x, r_x)$ centred at x is given. Then there are points $x_1, x_2, \ldots \in E$ such that the cubes $Q(x_j, r_{x_j})$ cover E and every point of $\mathbb{R}^d$ lies in at most N_d of them, where N_d depends only on d.*

Proof. Assume that E is nonempty and let $D := \sup_{x,y\in E} |x-y|_\infty < \infty$. If $r_x \geq 2D$ for some $x \in E$, then $E \subseteq Q(x, r_x)$ and this single cube suffices. Hence assume that $r_x < 2D$ for every $x \in E$.

Let $s_0 := \sup_{x\in E} r_x$ and choose $x_1 \in E$ with $r_{x_1} > s_0/2$. Given $x_1, \ldots, x_m$, let

$$s_m := \sup\Big\{ r_x : x \in E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j}) \Big\}$$

and choose $x_{m+1} \in E \setminus \bigcup_{j\leq m} Q(x_j, r_{x_j})$ with $r_{x_{m+1}} > s_m/2$. The selection stops if this set is empty. If $j < i$, then x_i lies outside $Q(x_j, r_{x_j})$, and x_i was a possible choice when x_j was chosen. Hence

$$|x_i - x_j|_\infty > \frac{r_{x_j}}{2} \qquad \text{and} \qquad r_{x_i} \leq s_{j-1} < 2r_{x_j}. \tag{6}$$

If the selection stops, the cubes $Q(x_j, r_{x_j})$ cover E by construction. Assume that it does not stop, and suppose, for contradiction, that some $x \in E$ is not covered. Then x is a possible choice at every step, so $r_x \leq s_{j-1} < 2r_{x_j}$ for every j. If $j < i$, then by (6)

$$\frac{r_{x_i}}{8} + \frac{r_{x_j}}{8} < \frac{3r_{x_j}}{8} < |x_i - x_j|_\infty,$$

so the cubes $Q(x_j, r_{x_j}/4)$ are pairwise disjoint. Since $x_j \in E$ and $r_{x_j} < 2D$, they are all contained in $Q(x_1, 4D)$. Comparing volumes, for every N,

$$N\left(\frac{r_x}{8}\right)^d < \sum_{j=1}^{N} \left(\frac{r_{x_j}}{4}\right)^d \leq (4D)^d,$$

which is false for large N. Hence every $x \in E$ is covered.

Fix $z \in \mathbb{R}^d$. For $\varepsilon \in \{-1, 1\}^d$ let

$$O_\varepsilon := \{y \in \mathbb{R}^d : \varepsilon_k(y_k - z_k) \geq 0 \text{ for } k = 1, \ldots, d\}.$$

Every $y \in \mathbb{R}^d$ lies in O_ε with $\varepsilon_k := 1$ if $y_k \geq z_k$ and $\varepsilon_k := -1$ otherwise, so these 2^d orthants cover $\mathbb{R}^d$. If $y, y' \in O_\varepsilon$, then $y_k - z_k$ and $y'_k - z_k$ have the same sign for every k, and hence

$$|y - y'|_\infty \leq \max\{|y - z|_\infty,\ |y' - z|_\infty\}. \tag{7}$$

For each ε let

$$J_\varepsilon := \{x_i : x_i \in O_\varepsilon \text{ and } z \in Q(x_i, r_{x_i})\}.$$

Every cube $Q(x_i, r_{x_i})$ containing z has its centre in some O_ε, so the number of such cubes is at most the total number of points in the sets J_ε.

Fix ε with J_ε nonempty and let $j := \min\{i : x_i \in J_\varepsilon\}$, so that x_j is the point of J_ε that was chosen first. Let n be the number of points in $J_\varepsilon \setminus \{x_j\}$, so that J_ε has $1 + n$ points. Let $x_i \in J_\varepsilon \setminus \{x_j\}$, so that $i > j$. Since z lies in both $Q(x_i, r_{x_i})$ and $Q(x_j, r_{x_j})$, (6) and (7) give

$$\frac{r_{x_j}}{2} < |x_i - x_j|_\infty \leq \max\{|x_i - z|_\infty,\ |x_j - z|_\infty\} \leq \max\Big\{\frac{r_{x_i}}{2},\ \frac{r_{x_j}}{2}\Big\}.$$

Hence $r_{x_i} > r_{x_j}$, and by (6) also $|x_i - z|_\infty \leq r_{x_i}/2 < r_{x_j}$.

For each $x_i \in J_\varepsilon \setminus \{x_j\}$ consider the cube $Q(x_i, r_{x_j}/2)$. It has the same centre as $Q(x_i, r_{x_i})$, but the fixed side $r_{x_j}/2$. If $x_i, x_m \in J_\varepsilon \setminus \{x_j\}$ with $m < i$, then $r_{x_m} > r_{x_j}$, so (6) gives $|x_i - x_m|_\infty > r_{x_m}/2 > r_{x_j}/2$. Thus the centres of these n cubes are more than their side length apart, and the cubes are pairwise disjoint. Moreover, for $x_i \in J_\varepsilon \setminus \{x_j\}$ every $y \in Q(x_i, r_{x_j}/2)$ satisfies

$$|y - z|_\infty \le |y - x_i|_\infty + |x_i - z|_\infty < \frac{r_{x_j}}{4} + r_{x_j} = \frac{5r_{x_j}}{4},$$

so all of them are contained in $Q(z, 5r_{x_j}/2)$. Since they are disjoint, their total volume is at most the volume of $Q(z, 5r_{x_j}/2)$, that is,

$$n\left(\frac{r_{x_j}}{2}\right)^d \le \left(\frac{5r_{x_j}}{2}\right)^d = 5^d\left(\frac{r_{x_j}}{2}\right)^d.$$

Hence $n \le 5^d$ and J_ε has at most $1+5^d$ points. Summing over the 2^d orthants, z lies in at most $2^d(1+5^d)$ of the cubes $Q(x_i, r_{x_i})$, and the lemma holds with $N_d := 2^d(1+5^d)$. □

Lemma 5. *Let v be a weight with $v(Q) > 0$ for every cube Q, and let f be a measurable function such that $|f|\,v$ is locally integrable. Then $M_v^c f$ is measurable, and*

$$v\big(\{x \in \mathbb{R}^d : M_v^c f(x) > \lambda\}\big) \le \frac{N_d}{\lambda}\,\|f\|_{L^1_v}$$

for every $\lambda > 0$, where N_d is the constant in Lemma 4.

Proof. Fix $\lambda > 0$, and call a cube $Q(x, r)$ *good* if

$$\int_{Q(x,r)} |f(y)|\,v(y)\,dy > \lambda\, v(Q(x,r)). \tag{8}$$

By the definition of M_v^c, we have $M_v^c f(x) > \lambda$ if and only if $Q(x, r)$ is good for some $r > 0$. For fixed $r > 0$, both sides of (8) are continuous functions of x. Indeed, if $x' \to x$, then $\chi_{Q(x',r)} \to \chi_{Q(x,r)}$ outside the boundary of $Q(x, r)$, which has measure zero, and $\chi_{Q(x',r)} \le \chi_{Q(x,r+2)}$ whenever $|x' - x| \le 1$, so the claim follows by dominated convergence since $|f|\,v$ and v are integrable on $Q(x, r+2)$. Hence the set U_r of all x for which $Q(x, r)$ is good is open, and so is $\{M_v^c f > \lambda\} = \bigcup_{r>0} U_r$. Since $\lambda > 0$ was arbitrary and $M_v^c f \ge 0$, the function $M_v^c f$ is measurable.

For every positive integer k let $E_k := \{x : |x| < k \text{ and } M_v^c f(x) > \lambda\}$. For each $x \in E_k$ we choose $r_x > 0$ such that $Q(x, r_x)$ is good. Since E_k is bounded, Lemma 4 gives points $x_1, x_2, \ldots \in E_k$ such that the cubes $Q(x_j, r_{x_j})$ cover E_k and satisfy $\sum_j \chi_{Q(x_j, r_{x_j})} \le N_d$. Using (8) for each $Q(x_j, r_{x_j})$ and monotone convergence to interchange sum and integral, we obtain

$$\begin{aligned} v(E_k) &\le \sum_j v\big(Q(x_j, r_{x_j})\big) \le \frac{1}{\lambda}\sum_j \int_{Q(x_j, r_{x_j})} |f|\,v\,dy \\ &= \frac{1}{\lambda}\int_{\mathbb{R}^d} \Big(\sum_j \chi_{Q(x_j, r_{x_j})}\Big)|f|\,v\,dy \le \frac{N_d}{\lambda}\,\|f\|_{L^1_v}. \end{aligned}$$

Since $E_1 \subseteq E_2 \subseteq \cdots$ and $\bigcup_k E_k = \{M_v^c f > \lambda\}$, continuity of the measure $v(x)\,dx$ from below gives the claim. □

Lemma 6. *Let v be a weight with $v(Q) > 0$ for every cube Q, and let $1 < q < \infty$. Then there is a constant $C_{q,d}$, depending only on q and d, such that*

$$\|M_v^c f\|_{L_v^q} \le C_{q,d}\, \|f\|_{L_v^q}$$

for all $f \in L_v^q(\mathbb{R}^d)$.

Proof. If $f = f_1 + f_2$ with $f_1 \in L_v^1(\mathbb{R}^d)$ and $f_2 \in L_v^\infty(\mathbb{R}^d)$, then $\int_Q |f|\, v\, dy \le \|f_1\|_{L_v^1} + \|f_2\|_{L_v^\infty}\, v(Q) < \infty$ for every cube Q. Hence M_v^c is defined on $L_v^1(\mathbb{R}^d) + L_v^\infty(\mathbb{R}^d)$, and $M_v^c f$ is measurable by Lemma 5. Since $|f+h| \le |f| + |h|$, the operator M_v^c is sublinear.

By Lemma 5, $v(\{M_v^c f > \lambda\}) \le \frac{N_d}{\lambda}\|f\|_{L_v^1}$ for every $f \in L_v^1(\mathbb{R}^d)$ and $\lambda > 0$. Moreover, if $f \in L_v^\infty(\mathbb{R}^d)$, then $|f| \le \|f\|_{L_v^\infty}$ outside a set of v-measure zero, so every average in the definition of $M_v^c f$ is at most $\|f\|_{L_v^\infty}$, and $\|M_v^c f\|_{L_v^\infty} \le \|f\|_{L_v^\infty}$. The Marcinkiewicz interpolation theorem [3, Theorem A.6], applied on the measure space $(\mathbb{R}^d, v(x)\,dx)$ with exponents 1 and ∞, now gives the claim with a constant depending only on q, N_d and 1, and hence only on q and d. □

2.2 Proof of Theorem 1

Proof of Theorem 1. Let $1 < p < \infty$, let $w \in A_p$ and let $f \in L_w^p(\mathbb{R}^d)$. If $w = 0$ almost everywhere, both sides of the inequality vanish, so we may assume that w is not almost everywhere zero. By Lemma 3, $w > 0$ almost everywhere and σ is a weight with $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q. In particular, the operators M_w^c and M_σ^c are defined.

We first prove a pointwise estimate for Mf. Fix $x \in \mathbb{R}^d$ and $r > 0$, let $Q := Q(x,r)$, and let $3Q$ be as in Lemma 1. Multiplying and dividing by $w(Q)\,\sigma(3Q)^{p-1}$, we obtain

$$\begin{aligned}
\left(\frac{1}{|Q|}\int_Q |f(y)|\,dy\right)^{p-1} &= \frac{1}{|Q|^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1} \\
&= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p}\cdot\frac{|Q|}{w(Q)}\cdot\frac{1}{\sigma(3Q)^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1} \\
&= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p}\cdot\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(y)|\,dy\right)^{p-1}.
\end{aligned} \tag{9}$$

Since $w(Q) \le w(3Q)$ and $|3Q| = 3^d|Q|$, the first factor on the right-hand side satisfies

$$\frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p} \le 3^{dp}\left(\frac{1}{|3Q|}\int_{3Q} w(y)\,dy\right)\left(\frac{1}{|3Q|}\int_{3Q}\sigma(y)\,dy\right)^{p-1} \le 3^{dp}\,[w]_{A_p}.$$

For the second factor, let $y \in Q$. By Lemma 1, $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, so

$$\begin{aligned}
\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz &= \frac{1}{\sigma(3Q)}\int_Q |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz \\
&\le \frac{1}{\sigma(Q(y,2\ell(Q)))}\int_{Q(y,2\ell(Q))} |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz \le M_\sigma^c\big(|f|\sigma^{-1}\big)(y).
\end{aligned}$$

This holds for every $y \in Q$, and therefore

$$\begin{aligned}
\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} &= \frac{1}{w(Q)}\int_Q\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} dy \\
&\le \frac{1}{w(Q)}\int_Q \Big(M_\sigma^c\big(|f|\sigma^{-1}\big)(y)\Big)^{p-1} w(y)^{-1}\,w(y)\,dy \\
&\le M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x),
\end{aligned}$$

where the last inequality holds because $Q = Q(x,r)$. Inserting both bounds into (9), taking the supremum over $r > 0$ and applying Lemma 2, we arrive at

$$Mf(x) \le 2^d\, 3^{\frac{dp}{p-1}}\, [w]_{A_p}^{\frac{1}{p-1}} \Big(M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x)\Big)^{\frac{1}{p-1}}. \tag{10}$$

We now take norms. Since $1 - \frac{p}{p-1} = -\frac{1}{p-1}$, we have $w^{1-\frac{p}{p-1}} = \sigma$, and similarly $\sigma^{1-p} = w$. The second identity gives

$$\big\| |f|\sigma^{-1}\big\|_{L^p_\sigma}^p = \int_{\mathbb{R}^d} |f(x)|^p\, \sigma(x)^{-p}\, \sigma(x)\, dx = \int_{\mathbb{R}^d} |f(x)|^p\, w(x)\, dx = \|f\|_{L^p_w}^p,$$

so $|f|\sigma^{-1} \in L^p_\sigma(\mathbb{R}^d)$. Using the first identity and Lemma 6 with $v = \sigma$ and $q = p$, we obtain

$$\begin{aligned}\int_{\mathbb{R}^d} \Big(\big(M_\sigma^c(|f|\sigma^{-1})(x)\big)^{p-1} w(x)^{-1}\Big)^{\frac{p}{p-1}} w(x)\, dx &= \int_{\mathbb{R}^d} \big(M_\sigma^c(|f|\sigma^{-1})(x)\big)^p\, \sigma(x)\, dx \\ &\le C_{p,d}^p\, \|f\|_{L^p_w}^p,\end{aligned}$$

so the function $\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}$ lies in $L_w^{p/(p-1)}(\mathbb{R}^d)$. Finally, we raise (10) to the power p, integrate against w, and apply Lemma 6 with $v = w$ and $q = \frac{p}{p-1}$. Together with the previous estimate this gives

$$\|Mf\|_{L^p_w}^p \le \Big(2^d\, 3^{\frac{dp}{p-1}}\Big)^p\, [w]_{A_p}^{\frac{p}{p-1}}\, C_{\frac{p}{p-1},d}^{\frac{p}{p-1}}\, C_{p,d}^p\, \|f\|_{L^p_w}^p.$$

Taking p-th roots, we conclude that

$$\|Mf\|_{L^p_w} \le c_{p,d}\, [w]_{A_p}^{\frac{1}{p-1}}\, \|f\|_{L^p_w}, \qquad c_{p,d} := 2^d\, 3^{\frac{dp}{p-1}}\, C_{\frac{p}{p-1},d}^{\frac{1}{p-1}}\, C_{p,d},$$

and $c_{p,d}$ depends only on p and d. □

2.3 Sharpness of the exponent

Proof of Proposition 2. Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$, and suppose, for contradiction, that there is a constant c such that

$$\|Mf\|_{L^p_w} \le c\, [w]_{A_p}^{\alpha}\, \|f\|_{L^p_w} \tag{11}$$

for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$. For every weight w that is not almost everywhere zero and every cube Q, Hölder's inequality with exponents p and $\frac{p}{p-1}$ gives

$$1 = \frac{1}{|Q|}\int_Q w^{\frac{1}{p}}\, w^{-\frac{1}{p}}\, dy \le \left(\frac{1}{|Q|}\int_Q w\, dy\right)^{\frac{1}{p}} \left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\, dy\right)^{\frac{p-1}{p}},$$

so $[w]_{A_p} \ge 1$. Hence (11) remains true if α is replaced by $\max\{\alpha, 0\}$, and we may assume that $0 \le \alpha < \frac{1}{p-1}$.

For $0 < \delta < 1$ we consider the weight $w_\delta(x) := |x|^{(d-\delta)(p-1)}$, for which $w_\delta^{-\frac{1}{p-1}}(x) = |x|^{\delta-d}$. By polar coordinates,

$$\int_{\{|y|<r\}} |y|^{\delta-d}\, dy = \omega_d\, \frac{r^\delta}{\delta} \qquad \text{for every } r > 0, \tag{12}$$

where ω_d denotes the surface measure of the unit sphere in $\mathbb{R}^d$.

We first show that $w_\delta \in A_p$ with

$$[w_\delta]_{A_p} \le C\,\delta^{1-p}, \tag{13}$$

where C depends only on p and d. The function w_δ is continuous and positive outside the origin, so it is a weight that is not almost everywhere zero. Write $a := (d-\delta)(p-1) \ge 0$, and let $Q := Q(c,\ell)$ be a cube. Every $y \in Q$ satisfies $|y-c| \le \frac{\sqrt{d}}{2}\,\ell$.

Suppose first that $|c| \le \sqrt{d}\,\ell$. Then every $y \in Q$ satisfies $|y| \le R\ell$ with $R := \frac{3\sqrt{d}}{2}$. Since $a \ge 0$, this gives $w_\delta \le (R\ell)^a$ on Q, and by (12),

$$\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy \le \frac{1}{\ell^d}\int_{\{|y|<R\ell\}} |y|^{\delta-d}\,dy = \frac{\omega_d\,R^\delta\,\ell^{\delta-d}}{\delta}.$$

Since $a + (\delta-d)(p-1) = 0$, we obtain

$$\left(\frac{1}{|Q|}\int_Q w_\delta\,dy\right)\left(\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy\right)^{p-1} \le (R\ell)^a\left(\frac{\omega_d\,R^\delta\,\ell^{\delta-d}}{\delta}\right)^{p-1} = R^{d(p-1)}\,\omega_d^{p-1}\,\delta^{1-p}.$$

Suppose instead that $|c| > \sqrt{d}\,\ell$. Then $|y-c| < \frac{1}{2}|c|$, so $\frac{1}{2}|c| \le |y| \le \frac{3}{2}|c|$ for every $y \in Q$. Hence $w_\delta \le \left(\frac{3}{2}|c|\right)^a$ and $|y|^{\delta-d} \le \left(\frac{1}{2}|c|\right)^{\delta-d}$ on Q, and

$$\left(\frac{1}{|Q|}\int_Q w_\delta\,dy\right)\left(\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy\right)^{p-1} \le \left(\tfrac{3}{2}\right)^a 2^{(d-\delta)(p-1)} = 3^a \le 3^{d(p-1)} \le 3^{d(p-1)}\,\delta^{1-p}.$$

Taking the supremum over all cubes proves (13) with $C := \max\{R^{d(p-1)}\omega_d^{p-1},\, 3^{d(p-1)}\}$.

Next we construct a function for which M is large in $L^p_{w_\delta}$. Let $f_\delta(x) := |x|^{\delta-d}\chi_{Q(0,2)}(x)$. Then $f_\delta \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, and since $Q(0,2) \subseteq \{|y| < 2\sqrt{d}\}$, (12) gives

$$\|f_\delta\|^p_{L^p_{w_\delta}} = \int_{Q(0,2)} |y|^{(\delta-d)p}\,|y|^{(d-\delta)(p-1)}\,dy = \int_{Q(0,2)} |y|^{\delta-d}\,dy \le \omega_d\,\frac{(2\sqrt{d})^\delta}{\delta} < \infty.$$

Moreover $\|f_\delta\|_{L^p_{w_\delta}} > 0$, so $f_\delta \in L^p_{w_\delta}(\mathbb{R}^d)$ is not the zero element. Let $x \in Q(0,2)$ with $x \ne 0$, and put $r := |x|_\infty$, so that $0 < r \le |x|$. The cube $Q(0,2r)$ contains x, is contained in $Q(0,2)$, and contains the ball $\{|y| < r\}$. By the definition of M and (12),

$$Mf_\delta(x) \ge \frac{1}{(2r)^d}\int_{Q(0,2r)} |y|^{\delta-d}\,dy \ge \frac{1}{(2r)^d}\int_{\{|y|<r\}} |y|^{\delta-d}\,dy = \frac{\omega_d}{2^d\delta}\,r^{\delta-d} \ge \frac{\omega_d}{2^d\delta}\,|x|^{\delta-d},$$

where the last inequality holds because $\delta - d < 0$ and $r \le |x|$. Thus $Mf_\delta \ge \frac{\omega_d}{2^d\delta} f_\delta$ almost everywhere on $Q(0,2)$, and this also holds outside $Q(0,2)$, where $f_\delta = 0$. Consequently

$$\|Mf_\delta\|_{L^p_{w_\delta}} \ge \frac{\omega_d}{2^d\delta}\,\|f_\delta\|_{L^p_{w_\delta}}. \tag{14}$$

Finally we combine these estimates. Applying (11) to w_δ and f_δ, and using (14), (13) and $\alpha \ge 0$, we obtain

$$\frac{\omega_d}{2^d\delta}\,\|f_\delta\|_{L^p_{w_\delta}} \le \|Mf_\delta\|_{L^p_{w_\delta}} \le c\,[w_\delta]^\alpha_{A_p}\,\|f_\delta\|_{L^p_{w_\delta}} \le c\,C^\alpha\,\delta^{\alpha(1-p)}\,\|f_\delta\|_{L^p_{w_\delta}}.$$

Dividing by $\|f_\delta\|_{L^p_{w_\delta}} \in (0,\infty)$ and multiplying by δ gives

$$\frac{\omega_d}{2^d} \le c\,C^\alpha\,\delta^{1-\alpha(p-1)} \qquad \text{for every } 0 < \delta < 1.$$

Since $\alpha < \frac{1}{p-1}$, the exponent $1-\alpha(p-1)$ is positive, so the right-hand side tends to 0 as $\delta \to 0$, while the left-hand side is a positive constant independent of δ. This is a contradiction, and the proof is complete. $\square$

A Proof of Proposition 1

We follow the proof of [3, Theorem 1.7].

Proof. We first show that $w > 0$ almost everywhere. Fix a cube Q and a measurable set $S \subseteq Q$. For $x \in Q$ the cube Q is one of the cubes in the supremum defining $M\chi_S(x)$, so $M\chi_S(x) \geq |S|/|Q|$. Hence (1) applied to $f = \chi_S$ gives

$$\left(\frac{|S|}{|Q|}\right)^p w(Q) \leq C^p\, w(S).$$

Since w is not almost everywhere zero, every bounded set S is contained in a cube Q with $w(Q) > 0$, so $w(S) = 0$ implies $|S| = 0$. Applying this to the sets $\{w = 0\} \cap \{|x| < k\}$, $k = 1, 2, \ldots$, shows that $w > 0$ almost everywhere. In particular $w(Q) > 0$ for every cube Q.

Now fix a cube Q and $\varepsilon > 0$, and let $f := (w+\varepsilon)^{-\frac{1}{p-1}}\,\chi_Q$. Since $w \leq w + \varepsilon$,

$$(w+\varepsilon)^{-\frac{p}{p-1}}\, w \leq (w+\varepsilon)^{-\frac{p}{p-1}}\,(w+\varepsilon) = (w+\varepsilon)^{-\frac{1}{p-1}} \leq \varepsilon^{-\frac{1}{p-1}},$$

so $f \in L^p_w(\mathbb{R}^d)$. Moreover, for $x \in Q$ the cube Q is one of the cubes in the supremum defining $Mf(x)$, so $Mf(x) \geq \frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy$, and therefore

$$\begin{aligned}
\left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^p w(Q) &= \int_Q \left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^p w(x)\,dx \\
&\leq \int_Q (Mf)^p\, w\,dx \\
&\leq \|Mf\|^p_{L^p_w} \\
&\leq C^p\,\|f\|^p_{L^p_w} \\
&= C^p \int_Q (w+\varepsilon)^{-\frac{p}{p-1}}\, w\,dx \\
&\leq C^p \int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dx.
\end{aligned}$$

Since w is locally integrable, $w < \infty$ almost everywhere, so $(w+\varepsilon)^{-\frac{1}{p-1}} > 0$ almost everywhere and the integral on the right-hand side is positive. Dividing by it we obtain

$$\frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^{p-1} \leq C^p.$$

As $\varepsilon \to 0$, $(w+\varepsilon)^{-\frac{1}{p-1}}$ increases to $w^{-\frac{1}{p-1}}$. Hence

$$\begin{aligned}
&\frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dy\right)^{p-1} \\
&\quad= \frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q \lim_{\varepsilon\to 0}(w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^{p-1} \\
&\quad= \lim_{\varepsilon\to 0}\frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^{p-1} \qquad \text{(monotone convergence)} \\
&\quad\leq C^p,
\end{aligned}$$

which is (2). Since $w(Q) > 0$, this also shows that $\int_Q w^{-\frac{1}{p-1}}\,dy < \infty$. □

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